

PlantCare: Scan, Identify, and Grow

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Project Overview

PlantCare is a multi-screen iOS application that helps users identify plants and learn how to care for them. Users can photograph a plant using the iPhone camera or select an existing photograph from the photo gallery. The application sends the selected image to a live plant-identification API and displays the most likely plant species.

After identifying the plant, the application presents information about its normal characteristics and behavior, including its growth cycle, sunlight requirements, watering needs, preferred environment, and general care instructions. When supported by the API, the application may also display possible visible plant problems or conditions. These condition results will be presented as suggestions rather than guaranteed diagnoses.

Users can save an identified plant, its photograph, identification result, condition information, and care instructions to a personal “My Garden” collection. Saved plants will remain available after the application is completely closed and reopened.

The target users are beginner gardeners, houseplant owners, students, and people who want a convenient way to identify and manage their plants.

Main Application Screens

1. Scan Plant

The Scan Plant screen is the application’s starting screen. Users can choose between taking a new photograph with the iPhone camera and selecting an existing image from the photo gallery.

The screen will include instructions encouraging users to take a clear photograph of plants. After selecting an image, the user can preview it and press the identify plant button.

2. Identification Result

This screen displays the submitted photograph, the most likely plant name, scientific name, confidence score, and alternative matches returned by the API. Users can select the most appropriate result or scan another image when the identification confidence is low.

3. Plant Profile and Care

This screen displays detailed information about the identified plant, including its growth behaviour, watering requirements, sunlight needs, preferred environment, growth cycle, and general care instructions.

When condition recognition is supported, the screen may also show possible visible problems with an appropriate confidence score. The user can then press “Save to My Garden” to store the plant locally.

4. My Garden

My Garden displays all plants saved by the user. Each entry contains the scanned photograph, plant name, identification date, condition summary, and care information. Users can open a saved plant to review its details or remove it from the collection.

External API Integration

The application will use the [Pl@ntNet identification API](#) to analyze photographs. The API accepts plant images and returns JSON containing likely species, scientific names, confidence scores, and related images. Its [disease-identification service](#) may be used for supported plants and conditions.

After identification, the scientific name can be used to request additional care information from the [Perennial Plant API](#), including watering, sunlight, plant cycle, and care-guide data.

The application will use `URLSession`, asynchronous functions, and `Codable` models for networking and JSON parsing. It will show a loading animation while an image is being analyzed and provide clear feedback for unavailable networks, invalid photographs, missing information, unsuccessful identification, API limits, and server errors. API keys will be kept outside the submitted public source files.

Local Data Persistence

SwiftData will store each saved plant’s name, scientific name, identification confidence, condition summary, care information, photograph reference, and date added. Plant photographs selected from the camera or gallery will be copied into the application’s local storage.

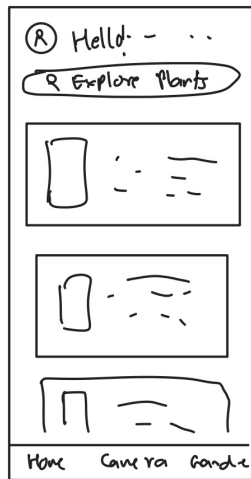
During the live demonstration, the team will scan a plant, save it to My Garden, close the application completely, and reopen it to prove that the plant remains stored.

Advanced User Interface

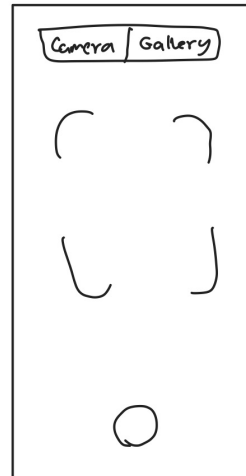
PlantCare will use SwiftUI and a nature-inspired color palette. Advanced interface elements will include a camera scanning overlay, animated loading indicator, confidence progress bars, smooth screen transitions, image cards, an animated confirmation when a plant is saved, and clear empty

and error states. The interface will support smooth scrolling and provide camera and photo-library permission feedback.

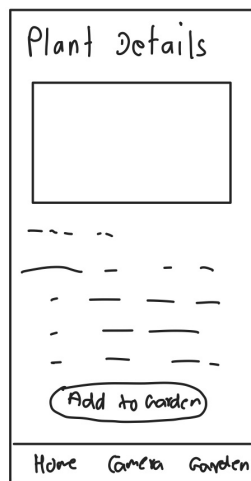
Home



Analysis



Plant



Garden

